

Pratham Arora

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Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India | July 2026

CGPA: 8.20 / 10.0

Research Experience

Undergraduate Researcher

Jan. 2026 – Present

Visual Benchmarking of VLMs | Sup. Prof. Pankaj Pansari | Plaksha University

- Constructed a 459-image benchmark with instrument-verified ground truth across 5 physical estimation tasks; personally handling all instrument measurements and building a 3-tier blur-degradation pipeline (OpenCV) to test model perceptual robustness.
- Benchmarked 6 VLMs (Gemini 3.1 Pro, GPT-5.4, Gemini-Robotics-ER, Claude Opus 4.7, Gemma 4, Qwen 3.5) across 3 independent runs each, finding performance collapses significantly under image blur across all frontier models.
- Identified a misidentification-vs-miscalibration failure dichotomy via top-10 error analysis per category: GPT-5.4 fails by misidentifying objects, while Gemini variants correctly identify but overestimate weights by 1.8–2.5×.

Undergraduate Researcher

June 2023 – Aug. 2023

Supervised by Prof. Sandeep Manjanna | Plaksha University

- Developed preprocessing pipeline (contrast adjustment, noise reduction, edge detection) to optimise **Segment Anything Model (SAM)** for agricultural crop-weed segmentation on sparse datasets, maximising zero-shot segmentation quality on out-of-distribution agricultural imagery.
- Improved batch processing throughput for **10,000+ image datasets** by implementing vectorized NumPy operations and Python multiprocessing.

Industry Experience

Freelance Web Developer

May 2026 – Present

PQRS Research (Dr. Niket Tandon) | Remote

- Leading a full redesign and feature buildout of the PQRS Research website in Next.js, including a dual onboarding system with distinct registration and intake flows for mentors and mentees.

AI Engineering Intern

June 2025 – July 2025

Cotality | Kolkata

- Architected a **FastAPI** backend with AST-based parsing for 15+ languages, enabling automated documentation generation across polyglot codebases at **50,000+ LOC/day**.
- Cut AI embedding costs by **85%** by building a hash-based system that fingerprints each function's AST and skips re-embedding unchanged code, eliminating redundant Azure OpenAI API calls.
- Built a **RAG pipeline** using LangChain and Azure OpenAI, backed by **Cosmos DB with IVF vector indexing** for semantic code retrieval — migrated from FAISS to support persistent, production-scale storage.
- Shipped **12 production REST API endpoints** with typed Pydantic schemas, powering real-time documentation preview in the client-facing web interface.

Freelance Web Developer

May 2025

Anuj Desai Associates | Remote

- Delivered a full-stack site with a public careers page and a **Firebase**-backed admin dashboard, giving firm partners a private portal to review and manage job applications.
- Secured admin access via **Google OAuth** with an email allowlist (Firebase Auth), enforcing partner-only access without a separate auth backend.
- Built a GitHub-editable content layer so the client can update site copy post-handoff without developer involvement.

Technical Projects

VR LLM Conversational Agent | Unity, Gemini API, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered a real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud STT/TTS, achieving average response latency of **1.8s** through parallel API calls, caching, and audio preprocessing.
- Designed **within-subjects experimental study** ($N = 18$) evaluating three latency feedback mechanisms (embodied gestures, visual indicators, verbal fillers) on perceived animacy and immersion via Godspeed questionnaire.
- Proved via **Wilcoxon signed-rank test** ($p < 0.05$) that embodied gestures significantly outperform visual loading indicators; **co-authored paper submitted to MIT's Presence journal** (under review).

AI Resume Builder & Interview Prep Tool | Next.js, Tailwind, Firebase, Gemini API

Jan. 2025 – Apr. 2025

- Built full-stack AI resume tool with real-time PDF preview and ATS-optimised generation using **Next.js SSR** and **Gemini 2.5 Flash** via streaming API.
- Integrated Gemini to generate personalised interview questions by cross-referencing uploaded resumes against a structured **DSA/STAR prep database**, surfacing role-specific preparation gaps.

- Implemented **Firebase Authentication** with Google OAuth for secure user management and session-based access control across the full application.

Kelp Forest Semantic Segmentation | *PyTorch, U-Net*

Jan. 2024 – May 2024

- Achieved **Top 100 finish (top 17%)** in a DrivenData satellite imagery competition using **U-Net with EfficientNet-B3** backbone for kelp forest segmentation in noisy real-world imagery.
- Built streak-detection pipeline using an ensemble of **CNN classifiers** (ResNet, DenseNet) with hard voting to filter atmospherically corrupted training images, achieving **F1 score of 0.952**.

Leadership & Involvement

Head of Technology

Apr. 2025 – Feb. 2026

Athleda – Plaksha Sports Society | Plaksha University

- Built and maintained a sports platform serving **500+ students**, shipping event registration, live-scoring, and photo galleries, while mentoring 2 junior developers through code reviews and production deployments.

LEAP.AI @ Plaksha

Feb. 2024 – Feb. 2025

Web Developer → Head of Web Development | Plaksha University

- Contributed as a core developer on a team of 7–8, then led the team through **2 full platform redesigns** as Head, growing the AI club site to **800+ monthly users** for events, workshops, and project showcases.

Skills

Languages: Python, SQL, JavaScript, TypeScript

AI/ML: PyTorch, LangChain, OpenCV, RAG Pipelines, Segment Anything (SAM)

Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, Pydantic, Tailwind CSS

Cloud & Tools: Azure (Cosmos DB, OpenAI), Firebase, FAISS, Git/GitHub, Vercel